1) a)

$z^3 = 8\mathrm{i}$

Using modulus-argument form: $8\mathrm{i} = 8\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 8 \Rightarrow r = 2$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2\mathrm{e}^{\frac{\pi}{6}\mathrm{i}} = \sqrt{3} + \mathrm{i}$

When $k = 1$: $z_1 = 2\mathrm{e}^{\frac{5\pi}{6}\mathrm{i}} = -\sqrt{3} + \mathrm{i}$

When $k = 2$: $z_2 = 2\mathrm{e}^{\frac{3\pi}{2}\mathrm{i}} = 2\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}} = -2\mathrm{i}$

$\therefore z = \sqrt{3} + \mathrm{i}$, $z = -\sqrt{3} + \mathrm{i}$ and $z = -2\mathrm{i}$

1) b)

$z^3 = -27$

Using modulus-argument form: $-27 = 27(\cos\pi + \mathrm{i}\sin\pi)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 27 \Rightarrow r = 3$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{3} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 3\mathrm{e}^{\frac{\pi}{3}\mathrm{i}} = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$

When $k = 1$: $z_1 = 3\mathrm{e}^{\pi\mathrm{i}} = -3$

When $k = 2$: $z_2 = 3\mathrm{e}^{\frac{5\pi}{3}\mathrm{i}} = 3\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

$\therefore z = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$, $z = -3$ and $z = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

1) c)

$z^2 = 1 + \sqrt{3}\mathrm{i}$

Using modulus-argument form: $1 + \sqrt{3}\mathrm{i} \ = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$

$r^2 = 2 \Rightarrow r = \sqrt{2}$

Since $2\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{6} + \frac{2k\pi}{2}$

When $k = 0$: $z_0 = \sqrt{2}\mathrm{e}^{\frac{\pi}{6}\mathrm{i}} = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$

When $k = 1$: $z_1 = \sqrt{2}\mathrm{e}^{\frac{7\pi}{6}\mathrm{i}} = \sqrt{2}\mathrm{e}^{-\frac{5\pi}{6}\mathrm{i}} = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$

$\therefore z = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$ and $z = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$

1) d)

$z^4 = -16$

Using modulus-argument form: $-16 = 16(\cos(-\pi) + \mathrm{i}\sin(-\pi))$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 16 \Rightarrow r = 2$

Since $4\theta = -\pi + 2k\pi$ for integer k: $\theta = -\frac{\pi}{4} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = 2\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}} = \sqrt{2} - \sqrt{2}\mathrm{i}$

When $k = 1$: $z_1 = 2\mathrm{e}^{\frac{\pi}{4}\mathrm{i}} = \sqrt{2} + \sqrt{2}\mathrm{i}$

When $k = 2$: $z_2 = 2\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}} = -\sqrt{2} + \sqrt{2}\mathrm{i}$

When $k = 3$: $z_3 = 2\mathrm{e}^{\frac{5\pi}{4}\mathrm{i}} = 2\mathrm{e}^{-\frac{3\pi}{4}\mathrm{i}} = -\sqrt{2} - \sqrt{2}\mathrm{i}$

$\therefore z = \sqrt{2} - \sqrt{2}\mathrm{i}$, $z = \sqrt{2} + \sqrt{2}\mathrm{i}$, $z = -\sqrt{2} + \sqrt{2}\mathrm{i}$ and $z = -\sqrt{2} - \sqrt{2}\mathrm{i}$

2) a)

$z^5 = 32\text{i}$

Using modulus-argument form: $32\text{i} = 32\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

$z^5 = r^5(\cos\theta + \text{i}\sin\theta)^5 = r^5(\cos 5\theta + \text{i}\sin 5\theta)$

$r^5 = 32 \Rightarrow r = 2$

Since $5\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{10} + \frac{2k\pi}{5}$

When $k = 0$: $z_0 = 2\left(\cos\frac{\pi}{10} + \text{i}\sin\frac{\pi}{10}\right)$

When $k = 1$: $z_1 = 2\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

When $k = 2$: $z_2 = 2\left(\cos\frac{9\pi}{10} + \text{i}\sin\frac{9\pi}{10}\right)$

When $k = 3$: $z_3 = 2\left(\cos\frac{13\pi}{10} + \text{i}\sin\frac{13\pi}{10}\right) = 2\left(\cos\left(-\frac{7\pi}{10}\right) + \text{i}\sin\left(-\frac{7\pi}{10}\right)\right)$

When $k = 4$: $z_4 = 2\left(\cos\frac{17\pi}{10} + \text{i}\sin\frac{17\pi}{10}\right) = 2\left(\cos\left(-\frac{3\pi}{10}\right) + \text{i}\sin\left(-\frac{3\pi}{10}\right)\right)$

$\therefore z = 2\left(\cos\frac{\pi}{10} + \text{i}\sin\frac{\pi}{10}\right)$, $z = 2\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$, $z = 2\left(\cos\frac{9\pi}{10} + \text{i}\sin\frac{9\pi}{10}\right)$,

$z = 2\left(\cos\left(-\frac{7\pi}{10}\right) + \text{i}\sin\left(-\frac{7\pi}{10}\right)\right)$ and $z = 2\left(\cos\left(-\frac{3\pi}{10}\right) + \text{i}\sin\left(-\frac{3\pi}{10}\right)\right)$

2) b)

$z^3 = 1 + \sqrt{3}\text{i}$

Using modulus-argument form: $1 + \sqrt{3}\text{i} = 2\left(\cos\frac{\pi}{3} + \text{i}\sin\frac{\pi}{3}\right)$

$z^3 = r^3(\cos\theta + \text{i}\sin\theta)^3 = r^3(\cos 3\theta + \text{i}\sin 3\theta)$

$r^3 = 2 \Rightarrow r = 2^{\frac{1}{3}}$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2^{\frac{1}{3}}\left(\cos\frac{\pi}{9} + \text{i}\sin\frac{\pi}{9}\right)$

When $k = 1$: $z_1 = 2^{\frac{1}{3}}\left(\cos\frac{7\pi}{9} + \text{i}\sin\frac{7\pi}{9}\right)$

When $k = 2$: $z_2 = 2^{\frac{1}{3}}\left(\cos\frac{13\pi}{9} + \text{i}\sin\frac{13\pi}{9}\right) = 2^{\frac{1}{3}}\left(\cos\left(-\frac{5\pi}{9}\right) + \text{i}\sin\left(-\frac{5\pi}{9}\right)\right)$

$\therefore z = 2^{\frac{1}{3}}\left(\cos\frac{\pi}{9} + \text{i}\sin\frac{\pi}{9}\right)$, $z = 2^{\frac{1}{3}}\left(\cos\frac{7\pi}{9} + \text{i}\sin\frac{7\pi}{9}\right)$

and $z = 2^{\frac{1}{3}}\left(\cos\left(-\frac{5\pi}{9}\right) + \text{i}\sin\left(-\frac{5\pi}{9}\right)\right)$

2) c)

$z^4 = -2\sqrt{2} + 2\sqrt{2}\mathrm{i}$

Using modulus-argument form: $-2\sqrt{2} + 2\sqrt{2}\mathrm{i} = 4\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 4 \Rightarrow r = 4^{\frac{1}{4}} = \sqrt{2}$

Since $4\theta = \frac{3\pi}{4} + 2k\pi$ for integer k: $\theta = \frac{3\pi}{16} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = \sqrt{2}\left(\cos\frac{3\pi}{16} + \mathrm{i}\sin\frac{3\pi}{16}\right)$

When $k = 1$: $z_1 = \sqrt{2}\left(\cos\frac{11\pi}{16} + \mathrm{i}\sin\frac{11\pi}{16}\right)$

When $k = 2$: $z_2 = \sqrt{2}\left(\cos\frac{19\pi}{16} + \mathrm{i}\sin\frac{19\pi}{16}\right) = \sqrt{2}\left(\cos\left(-\frac{13\pi}{16}\right) + \mathrm{i}\sin\left(-\frac{13\pi}{16}\right)\right)$

When $k = 3$: $z_3 = \sqrt{2}\left(\cos\frac{27\pi}{16} + \mathrm{i}\sin\frac{27\pi}{16}\right) = \sqrt{2}\left(\cos\left(-\frac{5\pi}{16}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{16}\right)\right)$

$\therefore z = \sqrt{2}\left(\cos\frac{3\pi}{16} + \mathrm{i}\sin\frac{3\pi}{16}\right)$, $z = \sqrt{2}\left(\cos\frac{11\pi}{16} + \mathrm{i}\sin\frac{11\pi}{16}\right)$,

$z = \sqrt{2}\left(\cos\left(-\frac{13\pi}{16}\right) + \mathrm{i}\sin\left(-\frac{13\pi}{16}\right)\right)$ and $z = \sqrt{2}\left(\cos\left(-\frac{5\pi}{16}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{16}\right)\right)$

2) d)

$z^8 = 8 + 8\sqrt{3}\text{i}$

Using modulus-argument form: $8 + 8\sqrt{3}\text{i} = 16\left(\cos\frac{\pi}{3} + \text{i}\sin\frac{\pi}{3}\right)$

$z^8 = r^8(\cos\theta + \text{i}\sin\theta)^8 = r^8(\cos 8\theta + \text{i}\sin 8\theta)$

$r^8 = 16 \Rightarrow r = 16^{\frac{1}{8}} = \sqrt{2}$

Since $8\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{24} + \frac{2k\pi}{8}$

When $k = 0$: $z_0 = \sqrt{2}\left(\cos\frac{\pi}{24} + \text{i}\sin\frac{\pi}{24}\right)$

When $k = 1$: $z_1 = \sqrt{2}\left(\cos\frac{7\pi}{24} + \text{i}\sin\frac{7\pi}{24}\right)$

When $k = 2$: $z_2 = \sqrt{2}\left(\cos\frac{13\pi}{24} + \text{i}\sin\frac{13\pi}{24}\right)$

When $k = 3$: $z_3 = \sqrt{2}\left(\cos\frac{19\pi}{24} + \text{i}\sin\frac{19\pi}{24}\right)$

When $k = 4$: $z_4 = \sqrt{2}\left(\cos\frac{25\pi}{24} + \text{i}\sin\frac{25\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{23\pi}{24}\right) + \text{i}\sin\left(-\frac{23\pi}{24}\right)\right)$

When $k = 5$: $z_5 = \sqrt{2}\left(\cos\frac{31\pi}{24} + \text{i}\sin\frac{31\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{17\pi}{24}\right) + \text{i}\sin\left(-\frac{17\pi}{24}\right)\right)$

When $k = 6$: $z_6 = \sqrt{2}\left(\cos\frac{37\pi}{24} + \text{i}\sin\frac{37\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{11\pi}{24}\right) + \text{i}\sin\left(-\frac{11\pi}{24}\right)\right)$

When $k = 7$: $z_7 = \sqrt{2}\left(\cos\frac{43\pi}{24} + \text{i}\sin\frac{43\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{5\pi}{24}\right) + \text{i}\sin\left(-\frac{5\pi}{24}\right)\right)$

$\therefore z = \sqrt{2}\left(\cos\frac{\pi}{24} + \text{i}\sin\frac{\pi}{24}\right)$, $z = \sqrt{2}\left(\cos\frac{7\pi}{24} + \text{i}\sin\frac{7\pi}{24}\right)$,

$z = \sqrt{2}\left(\cos\frac{13\pi}{24} + \text{i}\sin\frac{13\pi}{24}\right)$, $z = \sqrt{2}\left(\cos\frac{19\pi}{24} + \text{i}\sin\frac{19\pi}{24}\right)$

$z = \sqrt{2}\left(\cos\left(-\frac{23\pi}{24}\right) + \text{i}\sin\left(-\frac{23\pi}{24}\right)\right)$, $z = \sqrt{2}\left(\cos\left(-\frac{17\pi}{24}\right) + \text{i}\sin\left(-\frac{17\pi}{24}\right)\right)$,

$z = \sqrt{2}\left(\cos\left(-\frac{11\pi}{24}\right) + \text{i}\sin\left(-\frac{11\pi}{24}\right)\right)$ and $z = \sqrt{2}\left(\cos\left(-\frac{5\pi}{24}\right) + \text{i}\sin\left(-\frac{5\pi}{24}\right)\right)$

3) a)

$z^7 = 128\text{i}$

Using modulus-argument form: $128\text{i} = 128\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

$z^7 = r^7(\cos\theta + \text{i}\sin\theta)^7 = r^7(\cos 7\theta + \text{i}\sin 7\theta)$

$r^7 = 128 \Rightarrow r = 2$

Since $7\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{14} + \frac{2k\pi}{7}$

When $k = 0$: $z_0 = 2\text{e}^{\frac{\pi}{14}\text{i}}$

When $k = 1$: $z_1 = 2\text{e}^{\frac{5\pi}{14}\text{i}}$

When $k = 2$: $z_2 = 2\text{e}^{\frac{9\pi}{14}\text{i}}$

When $k = 3$: $z_3 = 2\text{e}^{\frac{13\pi}{14}\text{i}}$

When $k = 4$: $z_4 = 2\text{e}^{\frac{17\pi}{14}\text{i}} = 2\text{e}^{-\frac{11\pi}{14}\text{i}}$

When $k = 5$: $z_5 = 2\text{e}^{\frac{21\pi}{14}\text{i}} = 2\text{e}^{-\frac{\pi}{2}\text{i}}$

When $k = 6$: $z_6 = 2\text{e}^{\frac{25\pi}{14}\text{i}} = 2\text{e}^{-\frac{3\pi}{14}\text{i}}$

$\therefore z = 2\text{e}^{\frac{\pi}{14}\text{i}}$, $z = 2\text{e}^{\frac{5\pi}{14}\text{i}}$, $z = 2\text{e}^{\frac{9\pi}{14}\text{i}}$, $z = 2\text{e}^{\frac{13\pi}{14}\text{i}}$, $z = 2\text{e}^{-\frac{11\pi}{14}\text{i}}$, $z = 2\text{e}^{-\frac{\pi}{2}\text{i}}$ and $z = 2\text{e}^{-\frac{3\pi}{14}\text{i}}$

3) b)

$z^6 = -32 - 32\sqrt{3}\text{i}$

Using modulus-argument form: $-32 - 32\sqrt{3}\text{i} = 64\left(\cos\left(-\frac{2\pi}{3}\right) + \text{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

$z^6 = r^6(\cos\theta + \text{i}\sin\theta)^6 = r^6(\cos 6\theta + \text{i}\sin 6\theta)$

$r^6 = 64 \Rightarrow r = 2$

Since $6\theta = -\frac{2\pi}{3} + 2k\pi$ for integer k: $\theta = -\frac{2\pi}{18} + \frac{2k\pi}{6}$

When $k = 0$: $z_0 = 2\text{e}^{-\frac{\pi}{9}\text{i}}$

When $k = 1$: $z_1 = 2\text{e}^{\frac{2\pi}{9}\text{i}}$

When $k = 2$: $z_2 = 2\text{e}^{\frac{5\pi}{9}\text{i}}$

When $k = 3$: $z_3 = 2\text{e}^{\frac{8\pi}{9}\text{i}}$

When $k = 4$: $z_4 = 2\text{e}^{\frac{11\pi}{9}\text{i}} = 2\text{e}^{-\frac{7\pi}{9}\text{i}}$

When $k = 5$: $z_5 = 2\text{e}^{\frac{14\pi}{9}\text{i}} = 2\text{e}^{-\frac{4\pi}{9}\text{i}}$

$\therefore z = 2\text{e}^{-\frac{\pi}{9}\text{i}}$, $z = 2\text{e}^{\frac{2\pi}{9}\text{i}}$, $z = 2\text{e}^{\frac{5\pi}{9}\text{i}}$, $z = 2\text{e}^{\frac{8\pi}{9}\text{i}}$, $z = 2\text{e}^{-\frac{7\pi}{9}\text{i}}$ and $z = 2\text{e}^{-\frac{4\pi}{9}\text{i}}$

3) c)

$z^4 = \frac{9}{2} + \frac{9}{2}\sqrt{3}\mathrm{i}$

Using modulus-argument form: $\frac{9}{2} + \frac{9}{2}\sqrt{3}\mathrm{i} = 9\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 9 \Rightarrow r = \sqrt{3}$

Since $4\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{12} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = \sqrt{3}\mathrm{e}^{\frac{\pi}{12}\mathrm{i}}$

When $k = 1$: $z_1 = \sqrt{3}\mathrm{e}^{\frac{7\pi}{12}\mathrm{i}}$

When $k = 2$: $z_2 = \sqrt{3}\mathrm{e}^{\frac{13\pi}{12}\mathrm{i}} = \sqrt{3}\mathrm{e}^{-\frac{11\pi}{12}\mathrm{i}}$

When $k = 3$: $z_3 = \sqrt{3}\mathrm{e}^{\frac{19\pi}{12}\mathrm{i}} = \sqrt{3}\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}$

$\therefore z = \sqrt{3}\mathrm{e}^{\frac{\pi}{12}\mathrm{i}}$, $z = \sqrt{3}\mathrm{e}^{\frac{7\pi}{12}\mathrm{i}}$, $z = \sqrt{3}\mathrm{e}^{-\frac{11\pi}{12}\mathrm{i}}$ and $z = \sqrt{3}\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}$

4)

$z^3 = -2 + 2\mathrm{i}$

Using modulus-argument form: $-2 + 2\mathrm{i} = 2\sqrt{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 2\sqrt{2} \Rightarrow r = \sqrt{2}$

Since $3\theta = \frac{3\pi}{4} + 2k\pi$ for integer $k\theta = \frac{3\pi}{12} + \frac{2k\pi}{3}$:

When $k = 0$: $z_0 = \sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}} = 1 + \mathrm{i}$

When $k = 1$: $z_1 = \sqrt{2}\mathrm{e}^{\frac{11\pi}{12}\mathrm{i}} = -1.37 + 0.366\mathrm{i}$

When $k = 2$: $z_2 = \sqrt{2}\mathrm{e}^{\frac{19\pi}{12}\mathrm{i}} = \sqrt{2}\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}} = 0.366 - 1.37\mathrm{i}$

$\therefore z = 1 + \mathrm{i}$, $z = -1.37 + 0.366\mathrm{i}$ and $z = 0.366 - 1.37\mathrm{i}$

5) a)

$z^2 = -2 + 2\text{i}$

Using modulus-argument form: $-2 + 2\text{i} = 2\sqrt{2}\left(\cos\frac{3\pi}{4} + \text{i}\sin\frac{3\pi}{4}\right)$

$z^2 = r^2(\cos\theta + \text{i}\sin\theta)^2 = r^2(\cos 2\theta + \text{i}\sin 2\theta)$

$r^2 = 2\sqrt{2} \Rightarrow r = 2^{\frac{3}{4}}$

Since $2\theta = \frac{3\pi}{4} + 2k\pi$ for integer k: $\theta = \frac{3\pi}{8} + \frac{2k\pi}{2}$

When $k = 0$: $z_0 = 2^{\frac{3}{4}}\text{e}^{\frac{3\pi}{8}\text{i}} = 0.644 + 1.55\text{i}$

When $k = 1$: $z_1 = 2^{\frac{3}{4}}\text{e}^{\frac{11\pi}{8}\text{i}} = 2^{\frac{3}{4}}\text{e}^{-\frac{5\pi}{8}\text{i}} = -0.644 - 1.55\text{i}$

$\therefore z = 0.644 + 1.55\text{i}$ and $z = -0.644 - 1.55\text{i}$

5) b)

$z^2 = -2 + 4\text{i}$

Using modulus-argument form: $-2 + 4\text{i} = 2\sqrt{5}(\cos 2.0344 + \text{i}\sin 2.0344)$

$z^2 = r^2(\cos\theta + \text{i}\sin\theta)^2 = r^2(\cos 2\theta + \text{i}\sin 2\theta)$

$r^2 = 2\sqrt{5} \Rightarrow r = 2^{\frac{1}{2}}5^{\frac{1}{4}}$

Since $2\theta = 2.0344 + 2k\pi$ for integer k: $\theta = \frac{2.0344}{2} + \frac{2k\pi}{2}$

When $k = 0$: $z_0 = 2^{\frac{1}{2}}5^{\frac{1}{4}}(\cos 1.017 + \text{i}\sin 1.017) = 1.11 + 1.80\text{i}$

When $k = 1$: $z_1 = 2^{\frac{1}{2}}5^{\frac{1}{4}}(\cos 4.159 + \text{i}\sin 4.159) = -1.11 + 1.80\text{i}$

$\therefore z = 1.11 + 1.80\text{i}$ and $z = -1.11 + 1.80\text{i}$

5) c)

$z^4 = 4 - 4\mathrm{i}$

Using modulus-argument form: $4 - 4\mathrm{i} = 4\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 4\sqrt{2} \Rightarrow r = 2^{\frac{5}{8}}$

Since $4\theta = \left(-\frac{\pi}{4}\right) + 2k\pi$ for integer k: $\theta = -\frac{\pi}{16} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = 2^{\frac{5}{8}}\mathrm{e}^{-\frac{\pi}{16}\mathrm{i}} = 1.51 - 0.301\mathrm{i}$

When $k = 1$: $z_1 = 2^{\frac{5}{8}}\mathrm{e}^{\frac{7\pi}{16}\mathrm{i}} = 0.301 + 1.51\mathrm{i}$

When $k = 2$: $z_2 = 2^{\frac{5}{8}}\mathrm{e}^{\frac{15\pi}{16}\mathrm{i}} = -1.51 + 0.301\mathrm{i}$

When $k = 3$: $z = 2^{\frac{5}{8}}\mathrm{e}^{\frac{23\pi}{16}\mathrm{i}} = 2\mathrm{e}^{-\frac{7\pi}{16}\mathrm{i}} = -0.301 - 1.51\mathrm{i}$

$\therefore z = 1.51 - 0.301\mathrm{i}$, $z = 0.301 + 1.51\mathrm{i}$, $z = -1.51 + 0.301\mathrm{i}$ and $z = -0.301 - 1.51\mathrm{i}$

5) d)

$z^3 = 4 + 4\mathrm{i}$

Using modulus-argument form: $4 + 4\mathrm{i} = 4\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 4\sqrt{2} \Rightarrow r = 2^{\frac{5}{6}}$

Since $3\theta = \frac{\pi}{4} + 2k\pi$ for integer k: $\theta = \frac{\pi}{12} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2^{\frac{5}{6}}\mathrm{e}^{\frac{\pi}{12}\mathrm{i}} = 1.72 + 0.461\mathrm{i}$

When $k = 1$: $z_1 = 2^{\frac{5}{6}}\mathrm{e}^{\frac{9\pi}{12}\mathrm{i}} = -1.26 + 1.26\mathrm{i}$

When $k = 2$: $z_2 = 2^{\frac{5}{6}}\mathrm{e}^{\frac{17\pi}{12}\mathrm{i}} = 2^{\frac{5}{6}}\mathrm{e}^{-\frac{7\pi}{12}\mathrm{i}} = -0.461 + 1.72\mathrm{i}$

$\therefore z = 1.72 + 0.461\mathrm{i}$, $z = -1.26 + 1.26\mathrm{i}$ and $z = -0.461 + 1.72\mathrm{i}$

6)

Let $w = z - \mathrm{i}$ so that $w^2 = -4 + 12\mathrm{i}$

Using modulus-argument form: $-4 + 12\mathrm{i} \ = 4\sqrt{10}(\cos 1.8925 \ + \mathrm{i} \sin 1.8925)$

$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$

$r^2 = 4\sqrt{10} \Rightarrow r = 2 \times 10^{\frac{1}{4}}$

Since $2\theta = 1.8925 + 2k\pi$ for integer k: $\theta = \frac{1.8925}{2} + \frac{2k\pi}{2}$

When $k = 0$: $z_0 = 2 \times 10^{\frac{1}{4}}(\cos 0.946 \ + \mathrm{i}\sin 0.946) = 2.08 + 2.89\mathrm{i}$

When $k = 1$: $z_1 = 2 \times 10^{\frac{1}{4}}(\cos 4.088 \ + \mathrm{i}\sin 4.088) = -2.08 - 2.89\mathrm{i}$

Since $w = z - \mathrm{i}$ then $z = w + \mathrm{i}$

$\therefore z = 2.08 + 3.89\mathrm{i}$ and $z = -2.08 - 1.89\mathrm{i}$

7)

Let $w = z - 2$ so that $w^3 = 8\mathrm{i}$

Using modulus-argument form: $8\mathrm{i} \ = 8\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 8 \Rightarrow r = 2$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right) = \sqrt{3} + \mathrm{i}$

When $k = 1$: $z_1 = 2\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right) = -\sqrt{3} + \mathrm{i}$

When $k = 2$: $z_2 = 2\left(\cos\frac{9\pi}{6} + \mathrm{i}\sin\frac{9\pi}{6}\right) = -2\mathrm{i}$

Since $w = z - 2$ then $z = w + 2$

$\therefore z = 2 + \sqrt{3} + \mathrm{i}$, $z = 2 - \sqrt{3} + \mathrm{i}$ and $z = 2 - 2\mathrm{i}$

8) a)

$\left(-3\sqrt{3} + 3\mathrm{i}\right)^3 = 1\left(-3\sqrt{3}\right)^3 + 3\left(-3\sqrt{3}\right)^2(3\mathrm{i}) + 3\left(-3\sqrt{3}\right)(3\mathrm{i})^2 + (3\mathrm{i})^3 = 216\mathrm{i}$

$\therefore a = 0$ and $b = 216$

8) b)

$z^3 = 216\text{i}$

Using modulus-argument form: $216\text{i} = 216\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

$z^3 = r^3(\cos\theta + \text{i}\sin\theta)^3 = r^3(\cos 3\theta + \text{i}\sin 3\theta)$

$r^3 = 216 \Rightarrow r = 6$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer $k\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$:

When $k = 0$: $z_0 = 6\text{e}^{\frac{\pi}{6}\text{i}} = 3\sqrt{3} + 3\text{i}$

When $k = 1$: $z_1 = 6\text{e}^{\frac{5\pi}{6}\text{i}} = -3\sqrt{3} + 3\text{i}$

When $k = 2$: $z_2 = 6\text{e}^{\frac{9\pi}{6}\text{i}} = 6\text{e}^{-\frac{\pi}{2}\text{i}} = -6\text{i}$

$\therefore z = 3\sqrt{3} + 3\text{i}$, $z = -3\sqrt{3} + 3\text{i}$ and $z = -6\text{i}$

8) c)

$z^3 = 216\text{i}$ is not a polynomial with real coefficients.

Complex roots are in conjugate pairs only for polynomials with real coefficients.

9) a)

$(1 + \text{i})^4 = 1(1)^4 + 4(1)^3(\text{i}) + 6(1)^2(\text{i})^2 + 4(1)(\text{i})^3 + (\text{i})^4$

$= -4$

$\therefore a = -4$ and $b = 0$

9) b)

$z^4 = -4$

Using modulus-argument form: $-4 = 4(\cos\pi + \text{i}\sin\pi)$

$z^4 = r^4(\cos\theta + \text{i}\sin\theta)^4 = r^4(\cos 4\theta + \text{i}\sin 4\theta)$

$r^4 = 4 \Rightarrow r = \sqrt{2}$

Since $4\theta = \pi + 2k\pi$ for integer k: $\theta = \frac{\pi}{4} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = \sqrt{2}\left(\cos\frac{\pi}{4} + \text{i}\sin\frac{\pi}{4}\right) = 1 + \text{i}$

When $k = 1$: $z_1 = \sqrt{2}\left(\cos\frac{3\pi}{4} + \text{i}\sin\frac{3\pi}{4}\right) = -1 + \text{i}$

When $k = 2$: $z_2 = \sqrt{2}\left(\cos\frac{5\pi}{4} + \text{i}\sin\frac{5\pi}{4}\right) = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + \text{i}\sin\left(-\frac{3\pi}{4}\right)\right) = -1 - \text{i}$

When $k = 3$: $z_3 = \sqrt{2}\left(\cos\frac{7\pi}{4} + \text{i}\sin\frac{7\pi}{4}\right) = \sqrt{2}\left(\cos\left(-\frac{7\pi}{4}\right) + \text{i}\sin\left(-\frac{7\pi}{4}\right)\right) = 1 - \text{i}$

$\therefore z = 1 + \text{i}$, $z = -1 + \text{i}$, $z = -1 - \text{i}$ and $z = 1 - \text{i}$

9) c)

$z^4 = -4$ is a polynomial with real coefficients so complex roots come in conjugate pairs.

10) a)

$z^3 = 5 + 5\sqrt{3}\mathrm{i}$

Using modulus-argument form: $5 + 5\sqrt{3}\mathrm{i} = 10\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 10 \Rightarrow r = 10^{\frac{1}{3}}$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer $k\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$:

When $k = 0$: $z_0 = 10^{\frac{1}{3}}\mathrm{e}^{\frac{\pi}{9}\mathrm{i}}$

When $k = 1$: $z_1 = 10^{\frac{1}{3}}\mathrm{e}^{\frac{7\pi}{9}\mathrm{i}}$

When $k = 2$: $z_2 = 10^{\frac{1}{3}}\mathrm{e}^{\frac{13\pi}{9}\mathrm{i}} = 10^{\frac{1}{3}}\mathrm{e}^{-\frac{5\pi}{9}\mathrm{i}}$

$\therefore z = 10^{\frac{1}{3}}\mathrm{e}^{\frac{\pi}{9}\mathrm{i}}$, $z = 10^{\frac{1}{3}}\mathrm{e}^{\frac{7\pi}{9}\mathrm{i}}$ and $z = 10^{\frac{1}{3}}\mathrm{e}^{-\frac{5\pi}{9}\mathrm{i}}$

10) b)

$\left(10^{\frac{1}{3}}\mathrm{e}^{\frac{\pi}{9}\mathrm{i}}\right)^9 = 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000$

$\left(10^{\frac{1}{3}}\mathrm{e}^{\frac{7\pi}{9}\mathrm{i}}\right)^9 = 10^3\mathrm{e}^{7\pi\mathrm{i}} = 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000$

$\left(10^{\frac{1}{3}}\mathrm{e}^{-\frac{5\pi}{9}\mathrm{i}}\right)^9 = 10^3\mathrm{e}^{-5\pi\mathrm{i}} = 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000$

$\therefore$ all the roots satisfy $w^9 + 1000 = 0$

11)

$z^8 = (2 + 2\mathrm{i})^4$

But $(2 + 2\mathrm{i})^4 = 1(2)^4 + 4(2)^3(2\mathrm{i}) + 6(2)^2(2\mathrm{i})^2 + 4(2)^1(2\mathrm{i})^3 + (2\mathrm{i})^4$

$= 1(2)^4 + 4(2)^3(2\mathrm{i}) + 6(2)^2(2\mathrm{i})^2 + 4(2)^1(2\mathrm{i})^3 + (2\mathrm{i})^4$

$= 16 + 64\mathrm{i} - 96 - 64\mathrm{i} + 16$

$= -64$

$z^8 = -64$

Using modulus-argument form: $-64 = 64(\cos\pi + \mathrm{i}\sin\pi)$

$z^8 = r^8(\cos\theta + \mathrm{i}\sin\theta)^8 = r^8(\cos 8\theta + \mathrm{i}\sin 8\theta)$

$r^8 = 64 \Rightarrow r = 2^{\frac{3}{4}}$

Since $8\theta = \pi + 2k\pi$ for integer $k\theta = \frac{\pi}{8} + \frac{2k\pi}{8}$:

When $k = 0$: $z_0 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{\pi}{8}\mathrm{i}}$

When $k = 1$: $z_1 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{3\pi}{8}\mathrm{i}}$

When $k = 2$: $z_2 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{5\pi}{8}\mathrm{i}}$

When $k = 3$: $z_3 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{7\pi}{8}\mathrm{i}}$

When $k = 4$: $z_4 = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{7\pi}{8}\mathrm{i}}$

When $k = 5$: $z_5 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{11\pi}{8}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{5\pi}{8}\mathrm{i}}$

When $k = 6$: $z_6 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{13\pi}{8}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{3\pi}{8}\mathrm{i}}$

When $k = 7$: $z_7 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{15\pi}{8}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{\pi}{8}\mathrm{i}}$

$\therefore z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{3\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{5\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{7\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{7\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{5\pi}{8}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{3\pi}{8}\mathrm{i}}$ and $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{\pi}{8}\mathrm{i}}$

12) a)

$1 + \mathrm{i} = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right) = \sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$

12) b)

$\sqrt{3} - \mathrm{i} = 2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right) = 2\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}}$

12) c)

$z^4 = (1+\mathrm{i})^2\left(\sqrt{3}-\mathrm{i}\right)^2 = \left(\sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}\right)^2\left(2\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}}\right)^2 = 2\mathrm{e}^{\frac{\pi}{2}\mathrm{i}} \times 4\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = 8\mathrm{e}^{\frac{\pi}{6}\mathrm{i}}$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 8 \Rightarrow r = 8^{\frac{1}{4}} = 2^{\frac{3}{4}}$

Since $4\theta = \frac{\pi}{6} + 2k\pi$ for integer $k\theta = \frac{\pi}{24} + \frac{2k\pi}{4}$:

When $k = 0$: $z_0 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{\pi}{24}\mathrm{i}}$

When $k = 1$: $z_1 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{13\pi}{24}\mathrm{i}}$

When $k = 2$: $z_2 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{25\pi}{24}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{23\pi}{24}\mathrm{i}}$

When $k = 3$: $z_3 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{37\pi}{24}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{11\pi}{24}\mathrm{i}}$

$\therefore z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{\pi}{24}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{13\pi}{24}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{23\pi}{24}\mathrm{i}}$ and $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{11\pi}{24}\mathrm{i}}$

13) a)

$5\sqrt{3} + 5\mathrm{i} = 10\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

13) b)

$z^5 = 10^5\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right) = 100000\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)$

$w^4 = 100000\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)$

$w^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 100000 \Rightarrow r = 10^{\frac{5}{4}}$

Since $4\theta = \frac{5\pi}{6} + 2k\pi$ for integer k: $\theta = \frac{5\pi}{24} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = 10^{\frac{5}{4}}\left(\cos\frac{5\pi}{24} + \mathrm{i}\sin\frac{5\pi}{24}\right)$

When $k = 1$: $z_1 = 10^{\frac{5}{4}}\left(\cos\frac{17\pi}{24} + \mathrm{i}\sin\frac{17\pi}{24}\right)$

When $k = 2$: $z_2 = 10^{\frac{5}{4}}\left(\cos\frac{29\pi}{24} + \mathrm{i}\sin\frac{29\pi}{24}\right) = 10^{\frac{5}{4}}\left(\cos\left(-\frac{19\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{19\pi}{24}\right)\right)$

When $k = 3$: $z_3 = 10^{\frac{5}{4}}\left(\cos\frac{41\pi}{24} + \mathrm{i}\sin\frac{41\pi}{24}\right) = 10^{\frac{5}{4}}\left(\cos\left(-\frac{7\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{24}\right)\right)$

$\therefore \mathrm{z} = 10^{\frac{5}{4}}\left(\cos\frac{5\pi}{24} + \mathrm{i}\sin\frac{5\pi}{24}\right)$, $\mathrm{z} = 10^{\frac{5}{4}}\left(\cos\frac{17\pi}{24} + \mathrm{i}\sin\frac{17\pi}{24}\right)$,

$\mathrm{z} = 10^{\frac{5}{4}}\left(\cos\left(-\frac{19\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{19\pi}{24}\right)\right)$ and $\mathrm{z} = 10^{\frac{5}{4}}\left(\cos\left(-\frac{7\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{24}\right)\right)$

14)

$z^{\frac{5}{3}} = -1 - \sqrt{3}\mathrm{i}$

$z^5 = \left(-1 - \sqrt{3}\mathrm{i}\right)^3 = 1(-1)^3 + 3(-1)^2\left(-\sqrt{3}\mathrm{i}\right) + 3(-1)\left(-\sqrt{3}\mathrm{i}\right)^2 + 1\left(-\sqrt{3}\mathrm{i}\right)^3$

$= -1 - 3\sqrt{3}\mathrm{i} + 9 + 3\sqrt{3}\mathrm{i}$

$= 8$

$\therefore z^5 = 8$

Using modulus-argument form: $8 = 8(\cos 0 + \mathrm{i}\sin 0)$

$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$

$r^5 = 8 \Rightarrow r = 8^{\frac{1}{5}} = 2^{\frac{3}{5}}$

Since $5\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{5}$

When $k = 0$: $z_0 = 2^{\frac{3}{5}}\mathrm{e}^{0\mathrm{i}}$

When $k = 1$: $z_1 = 2^{\frac{3}{5}}\mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$

When $k = 2$: $z_2 = 2^{\frac{3}{5}}\mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$

When $k = 3$: $z_3 = 2^{\frac{3}{5}}\mathrm{e}^{\frac{6\pi}{5}\mathrm{i}} = 2^{\frac{3}{5}}\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$

When $k = 4$: $z_4 = 2^{\frac{3}{5}}\mathrm{e}^{\frac{8\pi}{5}\mathrm{i}} = 2^{\frac{3}{5}}\mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$

$\therefore z = 2^{\frac{3}{5}}\mathrm{e}^{0\mathrm{i}}$, $z = 2^{\frac{3}{5}}\mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$, $z = 2^{\frac{3}{5}}\mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$, $z = 2^{\frac{3}{5}}\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$ and $z = 2^{\frac{3}{5}}\mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$

15)

$z^{\frac{4}{3}} = \sqrt{2} + \sqrt{2}\mathrm{i}$

$z^4 = \left(\sqrt{2} + \sqrt{2}\mathrm{i}\right)^3 = 1\left(\sqrt{2}\right)^3 + 3\left(\sqrt{2}\right)^2\left(\sqrt{2}\mathrm{i}\right) + 3\left(\sqrt{2}\right)\left(\sqrt{2}\mathrm{i}\right)^2 + 1\left(\sqrt{2}\mathrm{i}\right)^3$

$= 2\sqrt{2} + 6\sqrt{2}\mathrm{i} - 6\sqrt{2} - 2\sqrt{2}\mathrm{i}$

$= -4\sqrt{2} + 4\sqrt{2}\mathrm{i}$

$\therefore z^5 = -4\sqrt{2} + 4\sqrt{2}\mathrm{i}$

Using modulus-argument form: $-4\sqrt{2} + 4\sqrt{2}\mathrm{i} = 8\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 8 \Rightarrow r = 8^{\frac{1}{4}} = 2^{\frac{3}{4}}$

Since $4\theta = \frac{3\pi}{4} + 2k\pi$ for integer k: $\theta = \frac{3\pi}{16} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{3\pi}{16}\mathrm{i}}$

When $k = 1$: $z_1 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{11\pi}{16}\mathrm{i}}$

When $k = 2$: $z_2 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{19\pi}{16}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{13\pi}{16}\mathrm{i}}$

When $k = 3$: $z_3 = 2^{\frac{3}{4}}\mathrm{e}^{\frac{27\pi}{16}\mathrm{i}} = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{5\pi}{16}\mathrm{i}}$

$\therefore z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{3\pi}{16}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{\frac{11\pi}{16}\mathrm{i}}$, $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{13\pi}{16}\mathrm{i}}$ and $z = 2^{\frac{3}{4}}\mathrm{e}^{-\frac{5\pi}{16}\mathrm{i}}$

16) a)

$z_1 = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

16) b)

$z_2 = 2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

16) c)

$z_1{}^3 z_2{}^2 = 2^3\left(\cos\frac{3\pi}{3} + \mathrm{i}\sin\frac{3\pi}{3}\right) \times 2^2\left(\cos\left(-\frac{2\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{6}\right)\right)$

$= 8 \times 4\left(\cos\left(\pi - \frac{\pi}{3}\right) + \mathrm{i}\sin\left(\pi - \frac{\pi}{3}\right)\right)$

$= 32\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

$w^6 = 32\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

$w^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)$

$r^6 = 32 \Rightarrow r = 2^{\frac{5}{6}}$

Since $6\theta = \frac{2\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{2\pi}{18} + \frac{2k\pi}{6}$

When $k = 0$: $z_0 = 2^{\frac{5}{6}}\left(\cos\frac{\pi}{9} + \mathrm{i}\sin\frac{\pi}{9}\right)$

When $k = 1$: $z_1 = 2^{\frac{5}{6}}\left(\cos\frac{4\pi}{9} + \mathrm{i}\sin\frac{4\pi}{9}\right)$

When $k = 2$: $z_2 = 2^{\frac{5}{6}}\left(\cos\frac{7\pi}{9} + \mathrm{i}\sin\frac{7\pi}{9}\right)$

When $k = 3$: $z_3 = 2^{\frac{5}{6}}\left(\cos\frac{10\pi}{9} + \mathrm{i}\sin\frac{10\pi}{9}\right) = 2^{\frac{5}{6}}\left(\cos\left(-\frac{8\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{8\pi}{9}\right)\right)$

When $k = 4$: $z_4 = 2^{\frac{5}{6}}\left(\cos\frac{13\pi}{9} + \mathrm{i}\sin\frac{13\pi}{9}\right) = 2^{\frac{5}{6}}\left(\cos\left(-\frac{5\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{9}\right)\right)$

When $k = 5$: $z_5 = 2^{\frac{5}{6}}\left(\cos\frac{16\pi}{9} + \mathrm{i}\sin\frac{16\pi}{9}\right) = 2^{\frac{5}{6}}\left(\cos\left(-\frac{2\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{9}\right)\right)$

$\therefore w = 2^{\frac{5}{6}}\left(\cos\frac{\pi}{9} + \mathrm{i}\sin\frac{\pi}{9}\right)$, $w = 2^{\frac{5}{6}}\left(\cos\frac{4\pi}{9} + \mathrm{i}\sin\frac{4\pi}{9}\right)$, $w = 2^{\frac{5}{6}}\left(\cos\frac{7\pi}{9} + \mathrm{i}\sin\frac{7\pi}{9}\right)$,

$w = 2^{\frac{5}{6}}\left(\cos\left(-\frac{8\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{8\pi}{9}\right)\right)$, $w = 2^{\frac{5}{6}}\left(\cos\left(-\frac{5\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{9}\right)\right)$

and $w = 2^{\frac{5}{6}}\left(\cos\left(-\frac{2\pi}{9}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{9}\right)\right)$

17) a)

$(a-1)^3 = a^3 - 3a^2 + 3a - 1$

17) b)

$z^3 - 3z^2 + 3z - 9 = z^3 - 3z^2 + 3z - 1 - 8$

$= (z-1)^3 - 8$

Hence solving: $(z-1)^3 = 8$

Let $w = z - 1$ so solving $w^3 = 8$

Using modulus-argument form: $8 = 8(\cos 0 + \mathrm{i}\sin 0)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 8 \Rightarrow r = 2$

Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2\mathrm{e}^{0\mathrm{i}} = 2$

When $k = 1$: $z_1 = 2\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -1 + \sqrt{3}\mathrm{i}$

When $k = 2$: $z_2 = 2\mathrm{e}^{\frac{4\pi}{3}\mathrm{i}} = 2\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = -1 - \sqrt{3}\mathrm{i}$

Since $w = z - 1$ then $z = w + 1$

$\therefore z = 3$, $z = \sqrt{3}\mathrm{i}$ and $z = -\sqrt{3}\mathrm{i}$

18)

First spot that:

$(a-1)^6 = a^6 - 6a^5 + 15a^4 - 20a^3 + 15a^2 - 6a + 1$

And so:

$(z-1)^6 + 1 = 0 \Rightarrow (z-1)^6 = -1$

Let $w = z - 1$ so solving $w^6 = -1$

Using modulus-argument form: $-1 = 1(\cos\pi + \mathrm{i}\sin\pi)$

$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)$

$r^6 = 1 \Rightarrow r = 1$

Since $6\theta = \pi + 2k\pi$ for integer k: $\theta = \frac{\pi}{6} + \frac{2k\pi}{6}$

When $k = 0$: $z_0 = \mathrm{e}^{\frac{\pi}{6}\mathrm{i}} = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{3\pi}{6}\mathrm{i}} = \mathrm{i}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{5\pi}{6}\mathrm{i}} = -\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$

When $k = 3$: $z_3 = \mathrm{e}^{\frac{7\pi}{6}\mathrm{i}} = -\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$

When $k = 4$: $z_4 = \mathrm{e}^{\frac{9\pi}{6}\mathrm{i}} = -\mathrm{i}$

When $k = 5$: $z_5 = \mathrm{e}^{\frac{11\pi}{6}\mathrm{i}} = \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$

Since $w = z - 1$ then $z = w + 1$

$\therefore z = 1 + \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$, $z = 1 + \mathrm{i}$, $z = 1 - \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$, $z = 1 - \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$, $z = 1 - \mathrm{i}$ and $z = 1 + \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$

19) a)

$z = 1 + \mathrm{i} = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

So that $z^n = \left(\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right)^n = \left(\sqrt{2}\right)^n\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right)$

Therefore we need $\sin\frac{n\pi}{4} = 0 \Rightarrow \frac{n\pi}{4} = k\pi$ for some integer k so $n = 4k$

But also $\cos\frac{n\pi}{4} > 0$ so n can't be odd

$\therefore n = 8$

19) b)

$z^8 = \left(\sqrt{2}\right)^8\left(\cos\frac{8\pi}{4} + \mathrm{i}\sin\frac{8\pi}{4}\right) = 16$

19) c)

Since $8\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{8}$

$\sqrt{2}(\cos 0 + \mathrm{i}\sin 0) = \sqrt{2}$

$\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right) = 1 + \mathrm{i}$

$\sqrt{2}\left(\cos\frac{2\pi}{4} + \mathrm{i}\sin\frac{2\pi}{4}\right) = \sqrt{2}\mathrm{i}$

$\sqrt{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right) = -1 + \mathrm{i}$

$\sqrt{2}\left(\cos\frac{4\pi}{4} + \mathrm{i}\sin\frac{4\pi}{4}\right) = -\sqrt{2}$

$\sqrt{2}\left(\cos\frac{5\pi}{4} + \mathrm{i}\sin\frac{5\pi}{4}\right) = -1 - \mathrm{i}$

$\sqrt{2}\left(\cos\frac{6\pi}{4} + \mathrm{i}\sin\frac{6\pi}{4}\right) = -\sqrt{2}\mathrm{i}$

$\sqrt{2}\left(\cos\frac{7\pi}{4} + \mathrm{i}\sin\frac{7\pi}{4}\right) = 1 - \mathrm{i}$

$\mathrm{z} = \sqrt{2}$, $\mathrm{z} = 1 + \mathrm{i}$, $\mathrm{z} = \sqrt{2}\mathrm{i}$, $\mathrm{z} = -1 + \mathrm{i}$, $\mathrm{z} = -\sqrt{2}$, $\mathrm{z} = -1 - \mathrm{i}$, $\mathrm{z} = -\sqrt{2}\mathrm{i}$ and $\mathrm{z} = 1 - \mathrm{i}$

20) a)

$z = -7 + 7\sqrt{3}\mathrm{i} = 14\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

So that $z^n = \left(14\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)\right)^n = 14^n\left(\cos\frac{2n\pi}{3} + \mathrm{i}\sin\frac{2n\pi}{3}\right)$

Therefore we need $\sin\frac{2n\pi}{3} = 0 \Rightarrow \frac{2n\pi}{3} = k\pi$ for some integer k so $n = \frac{3}{2}k$

But also $\cos\frac{n\pi}{3} > 0$ so n can't be odd $\therefore k = 2$

$\therefore n = 3$

20) b)

$z^3 = 14^3(\cos 2\pi + \mathrm{i}\sin 2\pi) = 2744$

20) c)

$14\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -7 + 7\sqrt{3}\mathrm{i}$

$14\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -7 - 7\sqrt{3}\mathrm{i}$

$\therefore \mathrm{z} = 14$, $\mathrm{z} = -7 + 7\sqrt{3}\mathrm{i}$ and $\mathrm{z} = -7 - 7\sqrt{3}\mathrm{i}$

21) a)

$$z = \frac{2\sqrt{3} + 2\mathrm{i}}{\sqrt{3} - \mathrm{i}} = \frac{4\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)}{2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)} = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$$

$$z^n = \left(2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)\right)^n = 2^n\left(\cos\frac{n\pi}{3} + \mathrm{i}\sin\frac{n\pi}{3}\right)$$

Therefore we need $\sin\frac{n\pi}{3} = 0 \Rightarrow \frac{n\pi}{3} = k\pi$ for some integer k so $n = 3k$

But also $\cos\frac{n\pi}{3} > 0$ so n can't be odd $\therefore k = 2$

$\therefore n = 6$

21) b)

$$z^6 = 2^6\left(\cos\frac{6\pi}{3} + \mathrm{i}\sin\frac{6\pi}{3}\right) = 64$$

21) c)

$2(\cos 0 + \mathrm{i}\sin 0) = 2$

$2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right) = 1 + \sqrt{3}\mathrm{i}$

$2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -1 + \sqrt{3}\mathrm{i}$

$2(\cos\pi + \mathrm{i}\sin\pi) = -2$

$2\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -1 - \sqrt{3}\mathrm{i}$

$2\left(\cos\frac{5\pi}{3} + \mathrm{i}\sin\frac{5\pi}{3}\right) = 1 - \sqrt{3}\mathrm{i}$

$\therefore \mathrm{z} = 2$, $\mathrm{z} = 1 + \sqrt{3}\mathrm{i}$, $\mathrm{z} = -1 + \sqrt{3}\mathrm{i}$, $\mathrm{z} = -2$, $\mathrm{z} = -1 - \sqrt{3}\mathrm{i}$ and $\mathrm{z} = 1 - \sqrt{3}\mathrm{i}$

22)

$16z^4 = (z-\mathrm{i})^4 \Rightarrow \frac{z^4}{(z-\mathrm{i})^4} = 16 \Rightarrow \left(\frac{z-\mathrm{i}}{z}\right)^4 = 16$

Let $w = \frac{z-\mathrm{i}}{z}$ so that $w^4 = 16$

Using modulus-argument form: $16 = 16(\cos 0 + \mathrm{i}\sin 0)$

$w^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 16 \Rightarrow r = 2$

Since $4\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{4}$

So $w = 2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}}$ for $k = 0, 1, 2$ and 3

Substituting back in: $2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}}z = z - \mathrm{i}$

So $2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}}z - z = -\mathrm{i} \Rightarrow \left(2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}} - 1\right)z = -\mathrm{i}$

$\therefore z = -\frac{\mathrm{i}}{2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}}-1} = \frac{\mathrm{i}}{1-2\mathrm{e}^{\frac{k\pi}{2}\mathrm{i}}}$

When $k = 0$, $\mathrm{e}^0 = 1$ so $z_0 = \frac{\mathrm{i}}{1-2} = -\mathrm{i}$

When $k = 1$, $\mathrm{e}^{\frac{\pi}{2}\mathrm{i}} = \mathrm{i}$ so $z_1 = \frac{\mathrm{i}}{1-2\mathrm{i}} \times \frac{1+\mathrm{i}}{1+2\mathrm{i}} = \frac{\mathrm{i}-1}{1+4} = -\frac{2}{5} + \frac{1}{5}\mathrm{i}$

When $k = 2$, $\mathrm{e}^{\pi\mathrm{i}} = -1$ so $z_2 = \frac{\mathrm{i}}{1+2} = \frac{1}{3}\mathrm{i}$

When $k = 3$, $\mathrm{e}^{\frac{3\pi}{2}\mathrm{i}} = -\mathrm{i}$ so $z_3 = \frac{\mathrm{i}}{1+2\mathrm{i}} \times \frac{1-\mathrm{i}}{1-2\mathrm{i}} = \frac{\mathrm{i}+1}{1+4} = \frac{2}{5} + \frac{1}{5}\mathrm{i}$

So the solutions are:

$z = -\mathrm{i}$, $z = -\frac{2}{5} + \frac{1}{5}\mathrm{i}$, $z = \frac{1}{3}\mathrm{i}$ and $z = \frac{2}{5} + \frac{1}{5}\mathrm{i}$

23) a)

$z^3 = 1 + \sqrt{3}\mathrm{i}$

Using modulus-argument form: $1 + \sqrt{3}\mathrm{i} = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 2 \Rightarrow r = 2^{\frac{1}{3}}$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2^{\frac{1}{3}}\mathrm{e}^{\frac{\pi}{9}\mathrm{i}}$

When $k = 1$: $z_1 = 2^{\frac{1}{3}}\mathrm{e}^{\frac{7\pi}{9}\mathrm{i}}$

When $k = 2$: $z_2 = 2^{\frac{1}{3}}\mathrm{e}^{\frac{13\pi}{9}\mathrm{i}} = 2^{\frac{1}{3}}\mathrm{e}^{-\frac{5\pi}{9}\mathrm{i}}$

$\therefore z = 2^{\frac{1}{3}}\mathrm{e}^{\frac{\pi}{9}\mathrm{i}}$, $z = 2^{\frac{1}{3}}\mathrm{e}^{\frac{7\pi}{9}\mathrm{i}}$ and $z = 2^{\frac{1}{3}}\mathrm{e}^{-\frac{5\pi}{9}\mathrm{i}}$

23) b)

The sum of the three roots is zero:

$2^{\frac{1}{3}}e^{\frac{\pi}{9}i} + 2^{\frac{1}{3}}e^{\frac{7\pi}{9}i} + 2^{\frac{1}{3}}e^{-\frac{5\pi}{9}i} = 0$

$e^{\frac{\pi}{9}i} + e^{\frac{7\pi}{9}i} + e^{-\frac{5\pi}{9}i} = 0$

So that $\text{Re}\left(e^{\frac{\pi}{9}i} + e^{\frac{7\pi}{9}i} + e^{-\frac{5\pi}{9}i}\right) = 0$

$\cos\frac{\pi}{9} + \cos\frac{7\pi}{9} + \cos\left(-\frac{5\pi}{9}\right) = 0$

$\therefore \cos\frac{\pi}{9} + \cos\frac{5\pi}{9} + \cos\frac{7\pi}{9} = 0$ as req.

24) a)

$z^3 = 6\sqrt{3} + 6i = 6\left(\sqrt{3} + i\right)$

Using modulus-argument form: $\sqrt{3} + i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

So $6\left(\sqrt{3} + i\right) = 12\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

$z^3 = r^3(\cos\theta + i\sin\theta)^3 = r^3(\cos 3\theta + i\sin 3\theta)$

$r^3 = 12 \Rightarrow r = 12^{\frac{1}{3}}$

Since $3\theta = \frac{\pi}{6} + 2k\pi$ for integer k: $\theta = \frac{\pi}{18} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 12^{\frac{1}{3}}e^{\frac{\pi}{18}i}$

When $k = 1$: $z_1 = 12^{\frac{1}{3}}e^{\frac{13\pi}{18}i}$

When $k = 2$: $z_2 = 12^{\frac{1}{3}}e^{\frac{25\pi}{18}i} = 12^{\frac{1}{3}}e^{-\frac{11\pi}{18}i}$

$\therefore z = 12^{\frac{1}{3}}e^{\frac{\pi}{18}i}$, $z = 12^{\frac{1}{3}}e^{\frac{13\pi}{18}i}$ and $z = 12^{\frac{1}{3}}e^{-\frac{11\pi}{18}i}$

24) b)

The sum of the three roots is zero:

$12^{\frac{1}{3}}e^{\frac{\pi}{18}i} + 12^{\frac{1}{3}}e^{\frac{13\pi}{18}i} + 12^{\frac{1}{3}}e^{-\frac{11\pi}{18}i} = 0$

$e^{\frac{\pi}{18}i} + e^{\frac{13\pi}{18}i} + e^{-\frac{11\pi}{18}i} = 0$

So that $\text{Im}\left(e^{\frac{\pi}{18}i} + e^{\frac{13\pi}{18}i} + e^{-\frac{11\pi}{18}i}\right) = 0$

$\sin\frac{\pi}{18} + \sin\frac{13\pi}{18} + \sin\left(-\frac{11\pi}{18}\right) = 0$

$\therefore \sin\left(\frac{\pi}{18}\right) - \sin\left(\frac{11\pi}{18}\right) + \sin\left(\frac{13\pi}{18}\right) = 0$ as req.

25) a)

$z_1 = 1 + \sqrt{3}\mathrm{i} = 2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$

25) b)

$z_2 = 1 + \sqrt{3}\mathrm{i} = 2\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}$

25) c)

$z_1^6 + z_2^6 = \left(2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}\right)^6 + \left(2\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}\right)^6 = 2^6\mathrm{e}^{\frac{6\pi}{3}\mathrm{i}} + 2^6\mathrm{e}^{-\frac{6\pi}{3}\mathrm{i}} = 64\mathrm{e}^{0\mathrm{i}} + 64\mathrm{e}^{0\mathrm{i}} = 128$

and $z_1^6 - z_2^6 = 2^6\mathrm{e}^{\frac{6\pi}{3}\mathrm{i}} - 2^6\mathrm{e}^{-\frac{6\pi}{3}\mathrm{i}} = 64\mathrm{e}^{0\mathrm{i}} - 64\mathrm{e}^{0\mathrm{i}} = 0$

25) d)

$\omega = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, $\omega^2 = \mathrm{e}^{\frac{4\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$ and $z_1 z_2 = 4$

$z_1^6 + \omega {z_2}^6 + \omega^2 (z_1 z_2)^6 = 64 + 64\omega + 4096\omega^2$

$= 64 + 64\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right) + 4096\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)$

$= 64 - 32 + 32\sqrt{3}\mathrm{i} + -2048 - 2048\sqrt{3}\mathrm{i}$

$= -2016 - 2016\sqrt{3}\mathrm{i}$